

ATTENTION ECONOMICS IN SHORT-FORM VIDEO: Groove-Edited Music, Engagement, and Recommendation-Linked Consumption

By Nirushanth (Nirushanth Murugadas)

ORCID ID: <https://orcid.org/0009-0002-2440-2904>

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MANUSCRIPT STATUS: Evidence-frozen draft based on the authoritative rebuilt survey (N=224) and the verified 20-participant within-subject experiment. Core bibliography for central claims has been source-checked. The manuscript uses the reproducibly re-fitted v11 computational results and preserves the study's causal boundaries.

Abstract

Short-form video platforms increasingly combine highly salient audiovisual content with personalized recommendation systems. This study examines a narrower mechanism at the intersection of music psychology, human-computer interaction, and recommender systems: whether groove-edited short-form content is associated with stronger engagement intentions, and whether user-reported replay tendencies and recommendation exposure are associated with additional consumption. The study uses two complementary datasets. First, a 20-participant within-subject experiment compared four short-form conditions: a neutral control (A), the same visual with groove/funk music (B), groove/funk music paired with a high-status/celebrity visual (C), and a real-world viral-style edit combining rapid editing and groove music (D). Second, a rebuilt international online survey contains 224 raw responses collected through a Google Form and SurveySwap. In the experiment, the primary A→B contrast showed higher mean enjoyment (3.45→5.65), motivation (2.05→4.45), and intention to continue watching similar videos (2.95→4.65); paired Wilcoxon tests reported Holm-adjusted p-values of .0066, .00070, and .0182 respectively. In the survey, reported recommendation exposure was positively associated with watching more groove content than planned (Spearman ρ = .538, N=216), while perceived algorithm influence also showed a strong positive association with the same outcome (ρ = .577, N=216). A deterministic five-fold cross-validated logistic-regression analysis produced ROC-AUC values of .843 for a behaviour-only model, .843 for a perceived-algorithm model, and .891 for an extended model including reported recommendation exposure. A repeated five-fold sensitivity analysis (20 repeats) gave mean AUCs of .844, .841, and .897 respectively. These results are consistent with a human–algorithm feedback framework in which content-level engagement signals and personalized exposure may reinforce repeated consumption. They do not establish that platform algorithms caused the behavior, nor do they measure dopamine, neural activity, or production recommender-system weights. The contribution is therefore a testable computational and behavioral framework rather than a causal claim about proprietary platforms.

Keywords

attention economics; recommender systems; short-form video; music reward; groove music; repeated consumption; human–algorithm interaction; HCI; computational social science; algorithmic amplification

1. Introduction

Recommendation systems turn individual interaction into an input to future content selection. Industrial recommender architectures can combine large-scale candidate generation with learned ranking models, making personalization responsive to recent user actions. Covington, Adams, and Sargin describe YouTube as a two-stage system involving candidate generation and ranking, illustrating why user behavior is not merely an outcome of recommendation but part of the information environment in which later recommendations are produced. [1]

Music provides a particularly useful case for studying this interaction because reward and repetition are not purely metaphorical concepts. PET and related neuroimaging work has linked intensely pleasurable music with activity in reward- and emotion-related regions. Salimpoor et al. further reported anatomically distinct dopamine release during anticipation and peak emotional experience. A later double-blind pharmacological study directly manipulated dopaminergic synaptic availability and found that dopamine influenced music-evoked reward experiences. [2–4] These findings motivate a mechanistic background for studying musical engagement, but they do not imply that ordinary short-form consumption is a clinical addiction or that the present study measured dopamine.

Repeated exposure provides a second bridge. Experimental work has found that prior exposure can increase liking and recognition for melodies, and later naturalistic work found increases in liking across repeated listening, with familiarity emerging as an important predictor. Earworm research also shows that repeated exposure can influence involuntary musical imagery and related subjective responses. [5–8] The relevance to short-form video is not that repetition automatically increases liking without bound; rather, repetition can alter familiarity, memorability, and engagement, creating conditions under which a recommendation loop may become self-reinforcing.

The gap motivating this project is narrower than the broad claim that “music is addictive.” The literature reviewed for this project contains substantial evidence on music reward, familiarity, repetition, and involuntary musical imagery, but much less direct work connecting groove-edited short-form content with recommender-system exposure and repeated consumption. The project therefore asks a human–algorithm question rather than a purely neuroscientific one.

2. Research Question and Hypotheses

Primary research question: How do user engagement signals generated by groove-edited short-form videos interact with recommendation systems to amplify repetitive consumption?

Operational definition. In this study, “groove-edited” refers to short-form videos using the study’s selected groove/funk or slowed-reverb music edit, often presented with rhythmic or visually salient editing. This is a study-specific operationalization and is not proposed as a universal musicological definition of the groove/funk/slowed-reverb category.

Hypothesis	Operational test	Claim level
H1	A→B: same visual, control music vs groove/funk music; primary outcomes are enjoyment, motivation, and similar-video intention.	Within-session experimental difference
H2	Replay urge ↔ reported similar recommendations.	Associative
H3	Replay urge ↔ watching more groove content than planned.	Associative
H4	Perceived algorithm influence ↔ watching more groove content than	Associative/predictive

Hypothesis	Operational test	Claim level
	planned.	
H5	Replay frequency ↔ watching more groove content than planned.	Exploratory associative
H6	Searching the original song ↔ discovering new songs because of recommendations.	Associative
H7	Reported recommendation exposure ↔ watching more groove content than planned.	Associative
H8	The joint evidence is consistent with a testable human–algorithm feedback framework.	Framework claim; not causal proof

3. Related Work

3.1 Music reward and anticipation

Blood and Zatorre used PET to examine highly pleasurable music and observed activity in regions implicated in reward, motivation, emotion, and arousal, including the ventral striatum. Salimpoor et al. combined PET and fMRI to distinguish anticipation from peak musical pleasure, reporting a temporal and anatomical separation within the dopaminergic system. Ferreri et al. then used a double-blind, within-subject pharmacological design to show that dopamine manipulation altered music-evoked reward experiences. [2–4] The present project uses this literature only to motivate why musical structure and anticipation are plausible engagement-relevant features.

3.2 Repetition, familiarity, and involuntary musical imagery

Repeated exposure has long been studied as a determinant of musical liking and recognition. Szpunar, Schellenberg, and Pliner found exposure-related changes in liking and memory; Madison and Schiölde later reported monotonic increases in liking across repeated presentations in a naturalistic four-week listening experiment, with familiarity emerging as a strong explanatory variable. [5,6] An earworm study by Killingly and Lacherez experimentally varied exposure to novel choruses and found exposure-related changes in familiarity and the desire to sing along. [7] These findings support treating repetition, memory, and replay tendency as measurable behavioral constructs rather than assuming that “catchiness” is self-evident.

3.3 Recommender systems and the human–algorithm loop

Industrial recommender systems are designed to learn from user interactions and rank content accordingly. The YouTube system described by Covington et al. uses candidate generation followed by ranking and is explicitly designed to respond to the evolving corpus and recent user actions. [1] In the present study, the proprietary ranking function is unavailable and is not reconstructed. Instead, recommendation exposure and perceived algorithmic influence are treated as self-reported observable variables. The technical problem is thus framed as a measurement-and-modeling problem: whether these variables add explanatory or predictive information about continued consumption.

4. Methods

4.1 Study design

The study uses a mixed behavioral design with a controlled within-subject video experiment and an observational online survey. The two datasets are analyzed separately and then triangulated. The experiment

addresses content-level engagement differences; the survey addresses naturally occurring relationships among replay, recommendation exposure, and self-reported additional consumption.

4.2 Experiment

Condition	Visual	Music / editing	Interpretive role
A — Control	Neutral motivational walking/running clip; no celebrity	Calm instrumental; ~20 s	Baseline
B — Groove	Same visual as A	Groove/funk music; otherwise held constant	Primary music contrast
C — Groove + Celebrity	High-status/celebrity visual	Same groove music as B	Visual-context comparison
D — Real Viral	Short-form viral-style edit	Groove music + fast edits/zoom/glow/motion	Ecological package comparison

Twenty participant records (P01–P20) are currently represented in the experiment dataset. Each participant evaluated A, B, C, and D on the same nine questions: enjoyment (1–10), rewatch likelihood (five-level ordinal response), like/save/share intention, song-search intention, motivation (1–10), self-imagination (three-level response), and intention to continue watching similar videos (1–10). Voluntary consent was included in the protocol. The experiment is a small within-subject pilot and should not be interpreted as a population estimate.

The primary controlled contrast is A→B because the experiment protocol explicitly kept the visual constant and changed the music. B→C is treated as a visual-context contrast rather than a pure celebrity effect because the stimulus changes more than the presence of a person in isolation. C→D is treated as an ecological package comparison because multiple editing and content features change simultaneously. This distinction is essential to preserve internal validity.

4.3 Survey

The rebuilt survey dataset contains 224 raw rows. The source file contains 224 distinct timestamps and no exact duplicate rows in the current audit. Two age values are non-numeric (“Prefer not to say.” and “Arnav”) and are retained as missing for age-specific analysis rather than being replaced. Recruitment sources were Google Forms and SurveySwap and will be reported explicitly as part of the sampling procedure.

Survey items cover listening frequency, replay frequency, post-listening feeling, self-imagined role, temporary confidence, original-song search, replay urge, songs staying in the head, perceived work/study help and distraction, similar recommendation exposure, recommendation-based song discovery, watching more than planned, top recommendation platform, and perceived algorithmic influence. The analytic N therefore varies by item because of missing responses; N=224 is the raw dataset count, not a guarantee that every analysis uses all 224 observations.

4.4 Statistical analysis

Ordinal relationships were evaluated using Spearman rank correlation with pairwise complete cases. Six predefined screening relationships were evaluated and Holm-adjusted p-values were used to control the family-wise error rate for that screen. A listener-only sensitivity analysis excluded respondents who selected “Never” for groove/funk/slowed-reverb listening. This sensitivity analysis does not alter the primary dataset or delete any respondent record.

For the experiment, paired Wilcoxon signed-rank tests were used for the primary ordinal/continuous-like outcomes. Effect magnitudes were summarized with rank-biserial effect sizes. The experiment’s multiple pairwise contrasts are treated as secondary to the A→B comparison.

For the computational component, the target was binary unplanned additional consumption, defined as reporting “Often” or “Always” to the item asking whether recommendations make the participant watch more groove content than planned. Logistic regression was evaluated with 5-fold stratified cross-validation after feature standardization. Three specifications were compared: (1) behaviour-only: replay urge + listening frequency + reported similar recommendations; (2) perceived-algorithm: algorithm influence + replay urge + listening frequency; and (3) extended: algorithm influence + replay urge + listening frequency + reported recommendation exposure.

5. Results

5.1 Experiment results

The condition means show a clear descriptive gradient across the four stimulus packages. The primary A→B comparison increased mean enjoyment from 3.45 to 5.65, motivation from 2.05 to 4.45, and intention to continue watching similar videos from 2.95 to 4.65. The corresponding paired Wilcoxon tests yielded Holm-adjusted $p=.0066$, $.00070$, and $.0182$ respectively, with large rank-biserial effect estimates of approximately .84, 1.00, and .80 in the current transcription.

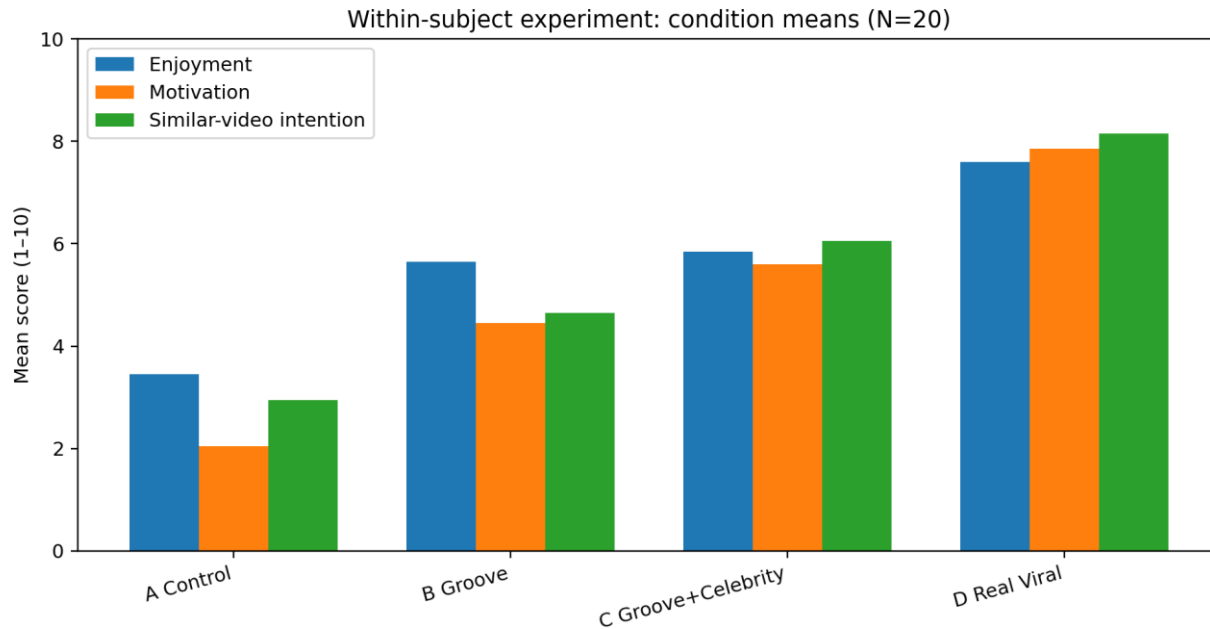


Figure 1. Mean outcome scores for the four experimental conditions. D is an ecological package comparison, not a single-variable causal manipulation.

The B→C comparison increased motivation from 4.45 to 5.60 and was statistically supported in the current pilot after Holm adjustment (adjusted $p=.0198$), whereas enjoyment (5.65→5.85) and similar-video intention (4.65→6.05) did not remain statistically supported after the same correction. C→D showed increases in all three primary outcomes, but this contrast combines multiple changes and is therefore descriptive rather than a clean causal estimate.

5.2 Survey association results

Relationship	N	Statistic	Holm-adjusted p	Interpretation
Replay urge ↔ similar	210	$\rho=.197$	.00844	Small positive association;

Relationship	N	Statistic	Holm-adjusted p	Interpretation
recommendations				weaker in listener-only sensitivity sample
Replay urge ↔ watch more than planned	210	$\rho=.351$	$<.000001$	Moderate positive association; robust in sensitivity analysis
Similar recommendations ↔ watch more than planned	216	$\rho=.538$	$<.000001$	Moderate/strong positive association; robust
Algorithm influence ↔ watch more than planned	216	$\rho=.577$	$<.000001$	Strong positive association; robust in sensitivity analysis
Replay frequency ↔ watch more than planned	210	$\rho=.159$	.02113	Weak association; not robust in listener-only sensitivity analysis
Song search ↔ recommendation discovery	179	$V=.257$	.00176	Small-to-moderate categorical association

The sensitivity analysis is informative because two relationships—replay urge with similar recommendations and replay frequency with extra consumption—attenuated after excluding non-listeners. By contrast, the associations involving reported recommendation exposure, perceived algorithm influence, and listening frequency remained statistically evident in the listener-only subset. This pattern argues for describing the strongest recommendation-related relationships as robust within the current sample while treating some replay relationships as more sample-sensitive.

5.3 Predictive modeling results

Model	N	Predictors	5-fold CV ROC-AUC	Interpretation
Behaviour-only	209	Replay urge + listening frequency + similar recommendations	.843	Behavioural variables carry substantial predictive signal.
Perceived-algorithm	209	Algorithm influence + replay urge + listening frequency	.843	Strong bivariate association, but no incremental CV gain over the behaviour-only baseline.
Extended	209	Algorithm influence + replay urge + listening frequency + similar recommendations	.891	Strongest exploratory discrimination; includes reported recommendation exposure.

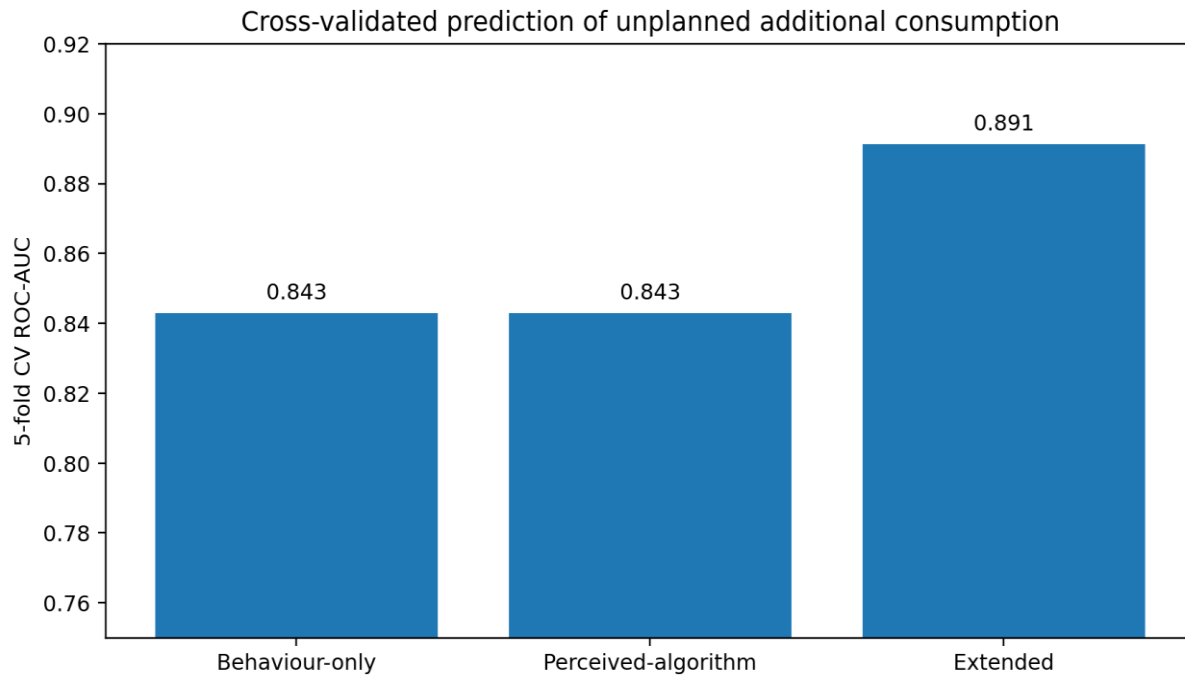


Figure 2. Cross-validated model discrimination for self-reported unplanned additional consumption.

The predictive results support a computational interpretation: the extended specification containing reported recommendation exposure provided the strongest discrimination of the self-reported outcome, whereas the perceived-algorithm specification did not improve discrimination over the behaviour-only baseline in this analysis. The result is not evidence that the actual production algorithm caused the behavior. It is also not a reconstruction of YouTube or Instagram ranking weights.

6. Integrated Human–Algorithm Feedback Framework

The evidence supports a testable framework rather than a completed causal graph. The proposed loop is:

Groove-edited content	Engagement intention	Replay / search behavior	Engagement signals	Recommendation exposure	More similar content	Further consumption
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The experiment primarily informs the left side of the framework: changing the music context while holding the visual constant changed reported enjoyment, motivation, and similar-video intention in the primary pilot contrast. The survey primarily informs the right and middle portions: recommendation exposure and perceived algorithm influence are associated with greater reported consumption, while replay and song-search variables connect individual engagement behavior with subsequent recommendation-related outcomes. The integrated interpretation is therefore that a feedback structure is plausible and worth testing with stronger longitudinal or platform-level data—not that this dataset has observed the full loop directly.

7. Discussion

7.1 What the study contributes

The strongest contribution is a change in problem framing. Instead of treating groove/funk short-form content as a purely musical preference or as evidence of “dopamine addiction,” the study treats content characteristics, user engagement, recommendation exposure, and repeated consumption as interacting components of a computational social system. The controlled A→B experiment provides a content-level

behavioral signal, while the survey provides a larger observational view of recommendation-related behavior. The computational model then tests whether these variables carry predictive information jointly.

7.2 Relationship to prior literature

The experiment is compatible with prior work showing that music reward, anticipation, familiarity, and repeated exposure can shape subjective response. [2–8] The survey results extend the question into the short-form platform context by linking user-reported engagement tendencies to reported recommendation exposure and additional consumption. The extension is conceptual and behavioral; it does not replicate neuroimaging findings or measure the platform’s internal objective function.

7.3 Why the strongest result is not “addiction”

The current evidence does not meet the evidentiary standard for a clinical addiction claim or for a causal claim about recommendation systems. “Watch more than planned” is a self-reported behavior indicator, not a diagnostic endpoint. Similarly, a perceived algorithm-influence score measures the participant’s perception or interpretation, not an observed ranking signal. The scientifically stronger statement is that recommendation-related variables are associated with self-reported additional consumption and improve prediction within this sample.

7.4 Implications for HCI and recommender-system research

A direct next step is to model the feedback structure longitudinally. The present data suggest candidate state variables—listening exposure, replay tendency, recommendation exposure, and unplanned additional consumption—that could be measured repeatedly over time. A future system-level study could combine controlled content interventions with instrumented viewing logs, allowing separation of content effects from recommendation effects and enabling causal mediation or sequential models. Such a design would directly test whether engagement signals produced by a content feature actually change future recommendation exposure.

8. Limitations

- Non-probability survey sampling. Recruitment through Google Forms and SurveySwap creates self-selection and limits generalizability. The sample should not be described as globally representative.
- Self-report. Recommendation exposure, algorithm influence, replay tendency, and additional consumption are reported by participants rather than retrieved from platform telemetry.
- No proprietary recommender data. The study does not observe candidate-generation, ranking, watch-time objectives, embedding representations, or platform-level recommendation weights.
- Small experiment. The within-subject experiment has $N=20$, so effect estimates should be interpreted as pilot evidence rather than precise population parameters.
- Stimulus confounding in C and D. $B \rightarrow C$ is a visual-context contrast and $C \rightarrow D$ changes several content features. Only $A \rightarrow B$ should be treated as the primary clean manipulation.
- Biological inference boundary. Dopamine and neural mechanisms are supported by prior literature, but no dopamine, neuroimaging, or physiological measurement was collected here.
- Core-source verification. The eight references used for the central manuscript claims were checked against publisher, PubMed, or DOI records; weaker media and commentary sources are not required for the core evidentiary argument.

9. Conclusion

This study provides preliminary evidence that groove-edited short-form content can alter immediate engagement intentions under a controlled within-subject contrast, and that recommendation-related variables are strongly associated with self-reported additional consumption in a larger observational sample. The computational models indicate that reported recommendation exposure adds predictive information in the extended specification, while the perceived-algorithm specification does not improve cross-validated discrimination over the behaviour-only baseline. Together, the findings are consistent with a human–algorithm feedback framework in which salient content generates engagement signals, engagement is followed by recommendation exposure, and recommendation exposure is associated with further consumption. The framework remains a hypothesis about system dynamics rather than a demonstrated causal mechanism. Stronger evidence would require longitudinal measurement, randomized content exposure, behavioral telemetry, and access to or controlled approximations of recommendation-system outputs.

10. Reproducibility and Data Governance

The project maintains a structured research workbook containing raw experiment transcription, coding keys, data-quality audits, statistical results, literature audits, model specifications, and publication gates. The current authoritative survey source is the rebuilt CSV dated 2026-08-30 with 224 raw rows. The experiment source is the 20-participant A/B/C/D dataset. Participant names are not used in the experiment dataset; participant codes P01–P20 are used instead. The manuscript intentionally distinguishes raw N from pairwise analytic N and does not fabricate missing demographic values.

Final analysis freeze. The authoritative survey contains 224 raw responses; pairwise analytic N varies by item completeness. The experiment contains 20 participants. The core manuscript statistics and model specification are frozen to the reproducible v11 analysis documented in the research workbook.

Model reproducibility audit. The final model figures reported here were independently re-fit from the 2026-08-30 rebuilt CSV under the stated StandardScaler + logistic-regression pipeline and deterministic 5-fold stratified cross-validation (random_state=42). A repeated 20×5-fold sensitivity analysis was used as a stability check.

11. Reference Set for Core Manuscript

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- [8] Freitas, C., Manzato, E., Burini, A., Taylor, M. J., Lerch, J. P., & Anagnostou, E. (2018). Neural Correlates of Familiarity in Music Listening: A Systematic Review and a Neuroimaging Meta-Analysis. *Frontiers in Neuroscience*, 12, 686. <https://doi.org/10.3389/fnins.2018.00686>

Appendix A. Core Literature Verification

The central scientific and technical references used by this manuscript were checked against publisher, PubMed, or DOI records. Verified core references are listed in the research workbook. Secondary media, blog, Reddit, YouTube, Wikipedia, and interview sources from the larger literature-note collection are not necessary for the manuscript’s core claims.

Appendix B. Publication-Freeze Checklist

Gate	Current state	Final interpretation / remaining venue work
Survey raw source	PASS	Use 224 raw rows; report pairwise analytic N.
Survey duplication audit	PASS	0 exact duplicate rows; 0 duplicate timestamps.
Experiment dataset	PASS — source-page verification confirmed by researcher	Freeze 20-participant pilot dataset; retain n=20 caveat.
Primary experiment contrast	PASS	A→B is the primary clean manipulation.
Core literature	PASS — 8 core references source-verified	Use the verified core reference set; unused secondary sources remain outside the core evidence layer.
Statistics	PASS — v11 statistics recomputed from authoritative CSV	Use v11 association and model results; preserve pairwise N.
Causality audit	PASS	Maintain association/predictive language; do not claim platform causality.
Submission package	SEPARATE VENUE CHECK	Journal-specific formatting and policy compliance are separate from research freeze.

Nirushanth (Nirushanth Murugadas)